

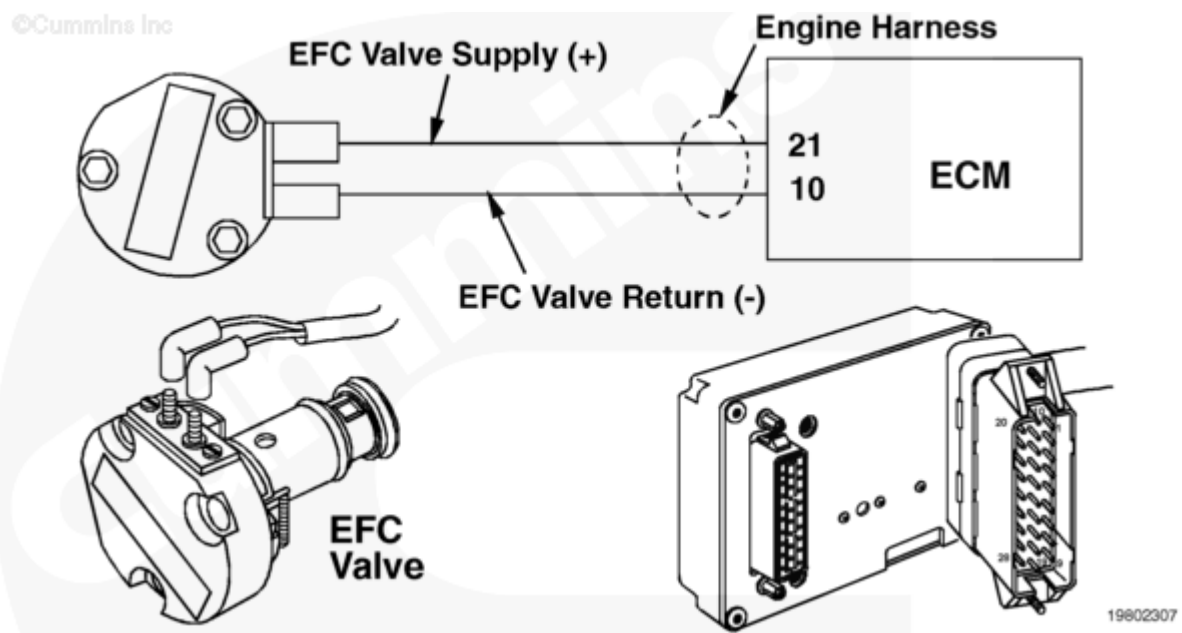
Fault Code: 511

Electronic Fuel Injection Control Valve Circuit - Voltage Below Normal or Shorted to Low Source

 Printable Version

Overview

Codes	Reason	Effect
Fault Code: 511 PID(P): S18 SPN: FMI: 4 Lamp: Flashing SRT: 000-643	Low current is detected at electronic fuel control valve primary return engine control module (ECM).	Engine may not start. Engine may shut down. Possible reduced engine performance. No overspeed protection



Electronic Fuel Control Valve Circuit

Circuit Description

The electronic fuel control valve is an actuator used by the ECM to control fuel flow to the fuel injectors. The electronic fuel control valve has two circuits: electronic fuel control valve supply, and electronic fuel control valve primary return.

Component Location

The electronic fuel control valve is bolted to the fuel pump on the left side of the engine.

Conditions For Running The Diagnostics

This diagnostic runs continuously when the keyswitch is in the ON position.

Conditions For Setting The Fault Codes

The Engine Control Module (ECM) detected the electronic fuel control valve supply was out of range low.

Action Taken When The Fault Code Is Active

- The ECM illuminates the red STOP ENGINE lamp immediately when the diagnostic runs and fails.

For Power Generation Applications:

- The generator set controller displays the fault immediately when the diagnostics runs and fails.

Conditions For Clearing The Fault Code

- Turn the keyswitch OFF. Allow the ECM to completely power down and turn the keyswitch ON.
- The fault code status displayed by the recommended Cummins® electronic service tool or equivalent will change to INACTIVE immediately after the diagnostic runs and passes.
- The ECM will turn off the red STOP ENGINE lamp immediately after the diagnostic runs and passes.
- The "Reset All Faults" command in the recommended Cummins® electronic service tool or equivalent can be used to clear active and inactive faults.

For Power Generation Applications:

- To validate the repair, start the engine and let it run for 1 minute at no load.
- The generator set controller will turn off the indicator immediately after the user presses reset.
- The "Reset All Faults" command in the recommended Cummins® electronic service tool or equivalent can be used to clear active faults.

Shoptalk

Possible causes of this fault code include:

- Malfunctioning or damaged engine wiring harness.
- Damaged or loose connectors.
- Malfunctioning or damaged electronic fuel control solenoid.

Refer to Troubleshooting Fault Code t05-511 (</qs3/pubsys2/xml/en/procedures/98/98-t05-511.html>)

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